

Ningzhe Shi

Ph.D. Candidate in Computer Science and Technology

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Research Profile

Ph.D. candidate researching 6G intelligent networks and AI service orchestration, with a focus on communication, sensing, computing, and knowledge resource optimization through reinforcement learning and optimization methods.

Education

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| <p>Ph.D. Institute of Computing Technology, Chinese Academy of Sciences; University of Chinese Academy of Sciences, Computer Science and Technology</p> <ul style="list-style-type: none"> • Ph.D. Advisor: Prof. Yiqing Zhou • Discipline rated A+ in the national subject assessment | <p>Beijing, China
Sept 2021 – present</p> |
| <p>B.Eng. Xidian University, Communication Engineering</p> <ul style="list-style-type: none"> • Undergraduate Research Advisor: Prof. Junyu Liu • Comprehensive recommendation ranking: 1st in class • Discipline rated A+ in the national subject assessment | <p>Xi'an, China
Sept 2017 – July 2021</p> |

Internship Experience

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| <p>China Mobile, Digital Intelligence Department, AI Agent Algorithm Intern</p> <ul style="list-style-type: none"> • Investigated long-horizon memory mechanisms for AI agents in multi-turn interactions, focusing on context loss, redundant historical information, memory evolution, and response consistency. • Surveyed retrieval-augmented memory, hierarchical storage, memory extraction, compression, forgetting, consolidation, and dynamic updating. • Reproduced and analyzed the Mem-alpha memory architecture and formulated memory maintenance as a policy-optimization problem. • Explored reinforcement-learning-based memory selection and updating. • Investigated memory-aware task scheduling under cloud-edge collaborative architectures, considering storage, retrieval, inference, communication latency, and response quality. | <p>China
Jan 2026 – Mar 2026</p> |
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Research Experience

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| <p>Institute of Computing Technology, Chinese Academy of Sciences, Technical Lead, Distributed System-Level Simulator for Large-Scale Full-Duplex Ad Hoc Networks</p> <ul style="list-style-type: none"> • Architected a WebSocket-based distributed simulation platform integrating a central controller, network-node workers, channel computation, mobility modeling, and visualization. • Designed priority-aware multi-channel access and cross-layer simulation logic covering traffic generation, routing, MAC access, physical transmission, interference, SINR, and reception decisions. • Built core algorithms and performance-evaluation workflows for large-scale scenarios involving more than 1,000 nodes. • Delivered a system prototype and one authorized software copyright. | <p>2022 – 2023</p> |
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Institute of Computing Technology, Chinese Academy of Sciences , Core Contributor, Knowledge-Aware Cloud-Edge Orchestration for RAG Services - Modeled heterogeneous knowledge bases, real-time computing loads, communication conditions, retrieval cost, generation quality, service delay, and energy consumption. - Designed a dynamic orchestration mechanism based on task semantics, node knowledge coverage, and resource states. - Built a cloud-edge RAG simulation environment for comparative evaluation.	2025 – 2026
Institute of Computing Technology, Chinese Academy of Sciences , Core Contributor, Memory-Aware Distributed Routing for Space AI Computing Networks - Modeled the coupling among task routing, node memory state, computing load, storage availability, and link conditions. - Designed a distributed routing method for high-dimensional and dynamically changing network states. - Evaluated the approach using a ground-based space-computing-network simulation environment.	2025 – 2026
Institute of Computing Technology, Chinese Academy of Sciences , Lead Researcher and First Author, ISAC-Driven AIGC Service Orchestration - Designed the content-accuracy-and-quality-aware metric and modeled sensing-computing-communication coupling for AIGC services. - Formulated the joint resource-allocation problem and designed an LP-guided hierarchical deep reinforcement learning framework. - Implemented comparative simulations, ablation studies, and performance analysis.	2024 – 2025
Institute of Computing Technology, Chinese Academy of Sciences , Task Leader, Technical Lead, and First Author, Deception-Based UAV-Assisted Secure Communications - Coordinated the technical route for secure communication modeling, deception signaling, UAV three-dimensional placement, beamforming, and power allocation. - Built simulation validation workflows and delivered paper and invention-patent outputs related to anti-interference and secure communication. - Achieved a gap of no more than 3% from the global optimum and improved secure transmission rate by more than 100% over the no-deception scheme.	2021 – 2023

Publications

Service Satisfaction Based User Selection and Resource Allocation for NOMA-Based Multi-Cell MEC Networks Ningzhe Shi , Yiqing Zhou, Ling Liu, Yihao Wu, Hanxiao Yu, Jinglin Shi 10.1109/TMC.2026.3673357 (IEEE Transactions on Mobile Computing)	2026
Content Accuracy and Quality Aware Resource Allocation Based on LP-Guided DRL for ISAC-Driven AIGC Networks Ningzhe Shi , Yiqing Zhou, Ling Liu, Jinglin Shi, Yihao Wu, Haiwei Shi, Hanxiao Yu 10.1109/TMC.2026.3652144 (IEEE Transactions on Mobile Computing)	2026
Long-Term Fairness From Real-Time Decisions: Reward Adaptive Multi-Agent DRL for Wireless Communication With Imperfect CSI Ningzhe Shi , Ling Liu, Yiqing Zhou, Hanxiao Yu, Yihao Wu, Jingya Yang, Jinglin Shi 10.1109/TCCN.2026.3692166 (IEEE Transactions on Cognitive Communications and Networking)	2026
Deception-Based Secure Communication for UAV-Assisted mmWave MIMO Systems Ningzhe Shi , Yiqing Zhou, Yu Zhang, Zhijun Han, Ling Liu, Jinglin Shi 10.23919/JCC.fa.2025-0337.202604 (China Communications)	2026

Intellectual Property

抗跟踪式干扰的信号发送方法、信号接收方法及系统 Granted invention patent: CN202211134854.X	Sept 2024
一种基于中继位置选择和功率分配的抗干扰方法及系统 Granted invention patent: CN202310078808.0	May 2025
无线网络资源分配系统的构建方法和资源管理方法 Granted invention patent: CN202310354794.0	Apr 2026
无线网络资源分配系统的构建方法和资源管理方法 Patent application: PCT/CN2023/090283	Apr 2023
一种全双工自组织网络实时仿真系统及方法 Patent application: CN202311719290.0	Dec 2023
一种多用户 MIMO 信号检测模型 Patent application: CN202410018551.4	Jan 2024
一种面向自组织网络的信道接入方法和通信系统 Patent application: CN202410348337.5	Mar 2024
一种移动自组网数据路由方法、工业自组网系统 Patent application: CN202510873622.3	June 2025
一种环境和资源状态感知的资源分配方法和系统 Patent application: CN202510955830.8	July 2025
基于集成感知与通信的 AI 生成内容服务的资源分配方法 Patent application: CN202511053332.0	July 2025
联合信道估计与信号检测的方法、信号接收机及存储介质 Patent application: CN202511536493.5	Oct 2025
一种面向高速移动通信的发射装置及语义传输方法 Patent application: CN202610406457.5	Mar 2026
大规模全双工网络仿真系统 Registered software copyright: 2024SR0082117	Jan 2024

Honors and Awards

Outstanding Graduate Student, Xinghang Program Journal of Electronics & Information Technology; national selection of 22 graduate students	June 2026
Outstanding Poster Award Frontiers in Electronics and Information Conference	June 2026
UCAS Outstanding Student Model University-wide top 1%	June 2026

ICT Outstanding Student Institute of Computing Technology / State Key Laboratory of Processors	Jan 2026
E Fund Financial Technology Doctoral Scholarship 12 recipients institute-wide	Jan 2025
UCAS Outstanding Student Cadre	May 2025
UCAS Outstanding Student	June 2023
UCAS First-Class Scholarship Awarded for five consecutive years	2021–2025
Outstanding Graduate, Xidian University	June 2021
First-Class Graduation Scholarship, Xidian University	June 2021
Honorable Mention, Mathematical Contest in Modeling	2020
National Scholarship	2019
National Second Prize, China Undergraduate Mathematical Contest in Modeling	2019
Third Prize, CETC-10 Communication Cup	2019

Academic Service

Program Committee: Technical Program Committee Member, IEEE International Conference on Communications (ICC), 2026

Academic Correspondence: Academic Correspondent, Digital Communications and Networks

Peer Review: IEEE Transactions on Mobile Computing (TMC); IEEE Transactions on Communications (TCOM); IEEE Internet of Things Journal; IEEE Transactions on Vehicular Technology (TVT); Science China Information Sciences; China Communications; IEEE International Conference on Communications (ICC); IEEE Vehicular Technology Conference (VTC); Journal on Communications; Journal of Electronics & Information Technology

Outreach and Public Engagement

Science Outreach

- Science outreach volunteer, Chinese Academy of Sciences Public Science Day
- Committee Member, UCAS Science Popularization Association
- Participated in public science communication, research demonstrations, and science outreach activities
- Outstanding Science Popularization Volunteer, Bureau of Academic Divisions, Chinese Academy of Sciences

Leadership and Student Service

Graduate Stage

- Vice President, Haidian Branch of the Volunteer Federation
- Committee Member, UCAS Science Popularization Association
- Class Mental Health Liaison

Undergraduate Stage

- Youth League Branch Secretary
- Team Leader, Horizon Volunteer Team, School of Telecommunications Engineering, Xidian University

Skills

Research and Modeling: Wireless and computing resource scheduling; integrated sensing, communication, computing, and intelligence; cloud-edge collaborative AI services; RAG service orchestration; AI Agent memory systems; space AI computing networks; cross-layer wireless network modeling

Optimization and Learning: Convex optimization; linear programming; evolutionary and heuristic algorithms; deep reinforcement learning; multi-agent reinforcement learning; generative diffusion models; quantum reinforcement learning; Lyapunov optimization

Programming and Simulation: Python; C++; MATLAB; Shell; PyTorch; custom reinforcement-learning environments; system-level network simulation; large-scale experiment automation; data processing and visualization

Systems and Engineering: Distributed simulation architecture; WebSocket-based module communication; wireless protocol-stack modeling; MAC and resource-allocation algorithm implementation; cloud-edge system modeling; performance evaluation and technical reporting

Research Project Execution: Research proposal preparation; technical scheme design; simulation and prototype validation; third-party testing support; mid-term project review; project completion and acceptance; patent and software-copyright preparation

Academic Communication: IEEE journal paper writing; English literature review; peer review; technical presentations; research reports; cross-team technical coordination